



University
Mohammed VI
Polytechnic



College of
Computing

Deliverable 3: Relational Algebra

Data Management Course

UM6P College of Computing

Professor: Karima Echihabi **Program:** Computer Engineering

Session: Fall 2026

Team Information

Team Name	PRINT("MySQL")
Member 1	YASSINE OUSSAMA
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Repository Link	https://github.com/abdelillahsaber/DatabaseGroupCC

1 Conversion

$$1. \quad \pi_{\text{firstName, lastName}} \left(\text{Person} \bowtie_{\text{Person.id} = \text{Reservation.madeByPid}} \text{Reservation} \right)$$

$$2. \quad \pi_{\text{pid}} \left(\text{Attached} \right) \cup \pi_{\text{pid}} \left(\sigma_{\text{role} = \text{'PI'}} \left(\text{Participates} \right) \right)$$

$$3. \quad \rho \left(A, \pi_{\text{labId}} \left(\sigma_{\text{building} = \text{'College of Computing'}} \left(\text{Laboratory} \right) \right) \right)$$

$$\rho \left(B, \pi_{\text{labId}} \left(\text{EquipmentUnit} \bowtie_{\sigma_{\text{category} = \text{'CPU Server'}}} \left(\text{EquipmentModel} \right) \right) \right)$$

$$A \cup B$$

$$4. \quad \rho \left(\text{CPU}, \pi_{\text{labId}} \left(\text{EquipmentUnit} \bowtie_{\sigma_{\text{category} = \text{'CPU Server'}}} \left(\text{EquipmentModel} \right) \right) \right)$$

$$\rho \left(\text{GPU}, \pi_{\text{labId}} \left(\text{EquipmentUnit} \bowtie_{\sigma_{\text{category} = \text{'GPU Server'}}} \left(\text{EquipmentModel} \right) \right) \right)$$

$$\text{CPU} \cap \text{GPU}$$

$$5. \rho(\text{Temp}, (\pi_{\text{labId}, \text{supervisorPid}}(\text{Laboratory})) \div (\pi_{\text{labId}}(\sigma_{\text{building} = \text{'College of Computing'}}(\text{Laboratory}))))$$

$$\pi_{\text{first_name}, \text{last_name}}(\text{Temp} \bowtie \text{Faculty})$$

$$6. \rho(\text{TempFundsLab}, \pi_{\text{budgetLine}}(\sigma_{\text{labId} = 2}(\text{FundsLab})))$$

$$\rho(\text{TempBudget}, \text{FundsPrj} \bowtie \text{TempFundsLab})$$

$$\rho(\text{TempResearchProject}, \pi_{\text{projectCode}}(\text{ResearchProject} \bowtie \text{TempBudget}))$$

$$\rho(\text{TempParticipate}, \pi_{\text{pid}, \text{projectCode}}(\text{Participates}))$$

$$\rho(\text{TempParticipateDivision}, \text{TempParticipate} \div \text{TempResearchProject})$$

$$\pi_{\text{pid}}(\text{TempParticipateDivision})$$

$$7. \rho(\text{reservationCount}, \gamma_{\text{pid}; \text{COUNT}(\text{resId}) \rightarrow \text{Total_Reservation}}(\text{Reservation}))$$

$$\rho(\text{P1}(1 \rightarrow \text{pid1}, 2 \rightarrow \text{TotalReservation1}), \text{reservationCount})$$

$$\rho(\text{P2}(1 \rightarrow \text{pid2}, 2 \rightarrow \text{TotalReservation2}), \text{reservationCount})$$

$$\pi_{\text{pid1}, \text{pid2}}(\sigma_{\text{Total_Reservation1} > \text{TotalReservation2}}(\text{P1} \times \text{P2}))$$

$$8. \rho(\text{Q}, \pi_{\text{pid}, \text{labId}}(\text{Person} \bowtie \text{Reservation} \bowtie \text{Reserves} \bowtie \text{Laboratory}))$$

$$\rho(\text{atLeastTwo}(1 \rightarrow \text{pid1}, 2 \rightarrow \text{labId1}, 3 \rightarrow \text{pid2}, 4 \rightarrow \text{labId2}), \text{Q} \times \text{Q})$$

$$\pi_{\text{pid1}}(\sigma_{\text{pid1} = \text{pid2} \wedge \text{labId1} \neq \text{labId2}}(\text{atLeastTwo}))$$

9.

$$\begin{aligned} & \pi_{\text{resId}} \left(\sigma_{\text{'2026-09-01'} \leq \text{plannedStart} < \text{'2026-10-01'}} \left(\right. \right. \\ & \quad \wedge \text{ name} = \text{'AI Movement'} \left. \right. \\ & \quad \text{Reserves} \bowtie \pi_{\text{serialNo}, \text{name}} \left(\pi_{\text{serialNo}, \text{labId}} (\text{EquipmentUnit}) \right. \\ & \quad \quad \left. \bowtie \pi_{\text{labId}, \text{name}} (\text{Laboratory}) \right) \\ & \quad \left. \left. \bowtie \pi_{\text{resId}, \text{plannedStart}} (\text{Reservation}) \right) \right) \end{aligned}$$

10.

$$\begin{aligned} & \pi_{\text{pid}} \left(\sigma_{m>1} \left(\right. \right. \\ & \quad \left. \gamma_{\text{pid}; \text{COUNT}(\text{projectCode}) \rightarrow m} \left(\sigma_{\text{role} = \text{'PI'}} (\text{Participates}) \right) \right) \left. \right) \end{aligned}$$

11.

$$\begin{aligned} & \rho \left(\text{R}, \pi_{\text{serialNo}} \left(\sigma_{m>1} \left(\gamma_{\text{serialNo}; \text{COUNT}(\text{labId}) \rightarrow m} (\text{EquipmentUnit}) \right) \right) \right) \\ & \quad \pi_{\text{approverPid}} \left(\pi_{\text{resId}, \text{approverPid}} (\text{Reservation}) \right. \\ & \quad \quad \left. \bowtie \pi_{\text{resId}} (\text{Reserves} \bowtie \text{R}) \right) \end{aligned}$$

12.

$$\begin{aligned} & \pi_{\text{madeByPid}} \left(\sigma_{\substack{\text{plannedStart} > \text{'2026-11-06'} \\ \vee \text{plannedEnd} < \text{'2026-11-06'}}} (\text{Reservation}) \right) \\ & \quad \cup \left(\pi_{\text{pid}} (\text{Person}) - \pi_{\text{madeByPid}} (\text{Reservation}) \right) \end{aligned}$$

13.

$$\rho\left(R, \gamma_{\text{labId}}; \text{COUNT}(\text{serialNo}) \rightarrow \text{count}(\text{EquipmentUnit})\right) \\ \pi_{\text{labId}}\left(\sigma_{\text{count} < R}\left(\left(\gamma_{\text{AVG}(\text{count}) \rightarrow \text{averageUni}(R)}\right) \times R\right)\right)$$

14.

$$\rho\left(R, \pi_{\text{approverPid}, \text{labId}}\left(\pi_{\text{approverPid}, \text{resId}}(\text{Reservation})\right.\right. \\ \left.\left.\bowtie\left(\pi_{\text{labId}, \text{serialNo}}(\text{EquipmentUnit}) \bowtie \text{Reserves}\right)\right)\right) \\ \rho\left(C, \gamma_{\text{approverPid}, \text{labId}}; \text{COUNT}(\ast) \rightarrow \text{count}(R)\right) \\ \rho\left(D, \gamma_{\text{labId}}; \text{MAX}(\text{count}) \rightarrow \text{max}(C)\right) \\ \pi_{\text{labId}, \text{approverPid}}\left(\sigma_{\text{count} = \text{max}(C \bowtie D)}\right)$$

15.

$$\pi_{\text{madeByPid}}\left(\sigma_{\text{'2025-01-01'} \leq \text{submissionTS} < \text{'2026-01-01'}}\left(\right.\right. \\ \left.\left.\pi_{\text{madeByPid}, \text{submissionTS}}(\text{Reservation})\right)\right)$$

2 Discussion

The only minor problem faced in this deliverable was the fact that the relational algebra expressions kept getting longer and longer as the questions went on. This issue was addressed simply by renaming intermediate results (using the ρ operator) to keep the expressions manageable.